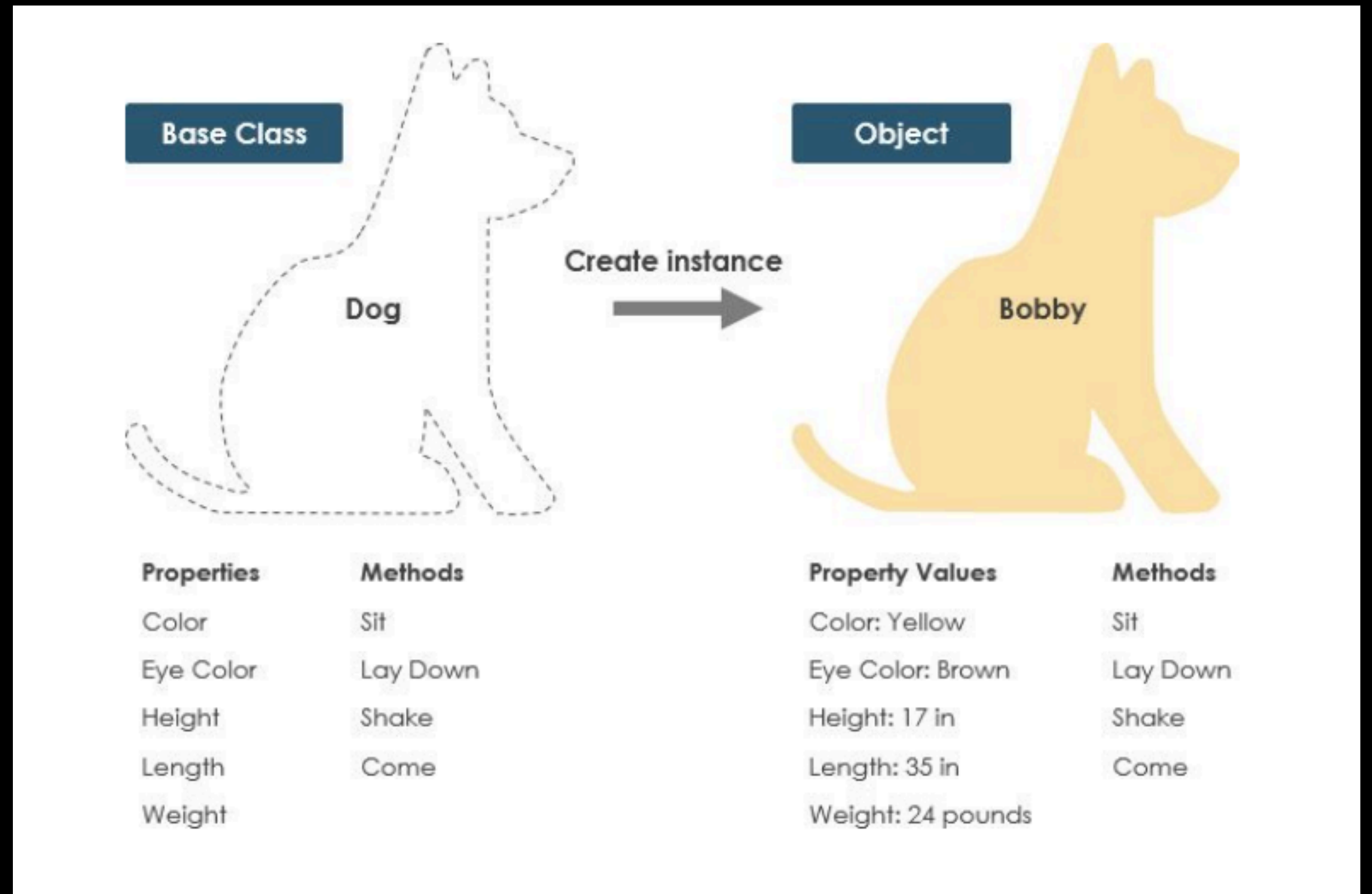


# M2S3 | UML Class Diagram

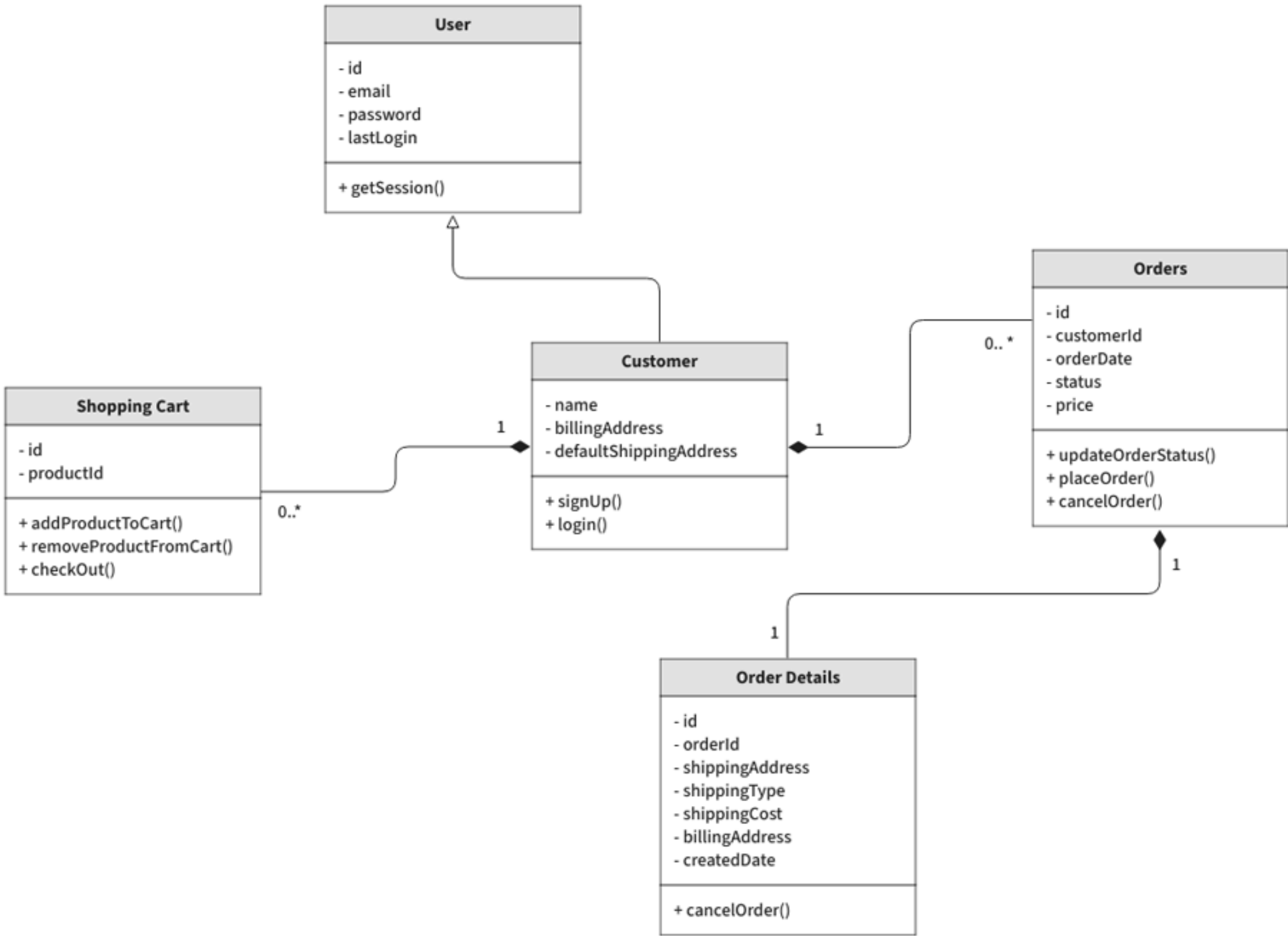
# Class and Object

why? instance creation?



# UML Class Diagram

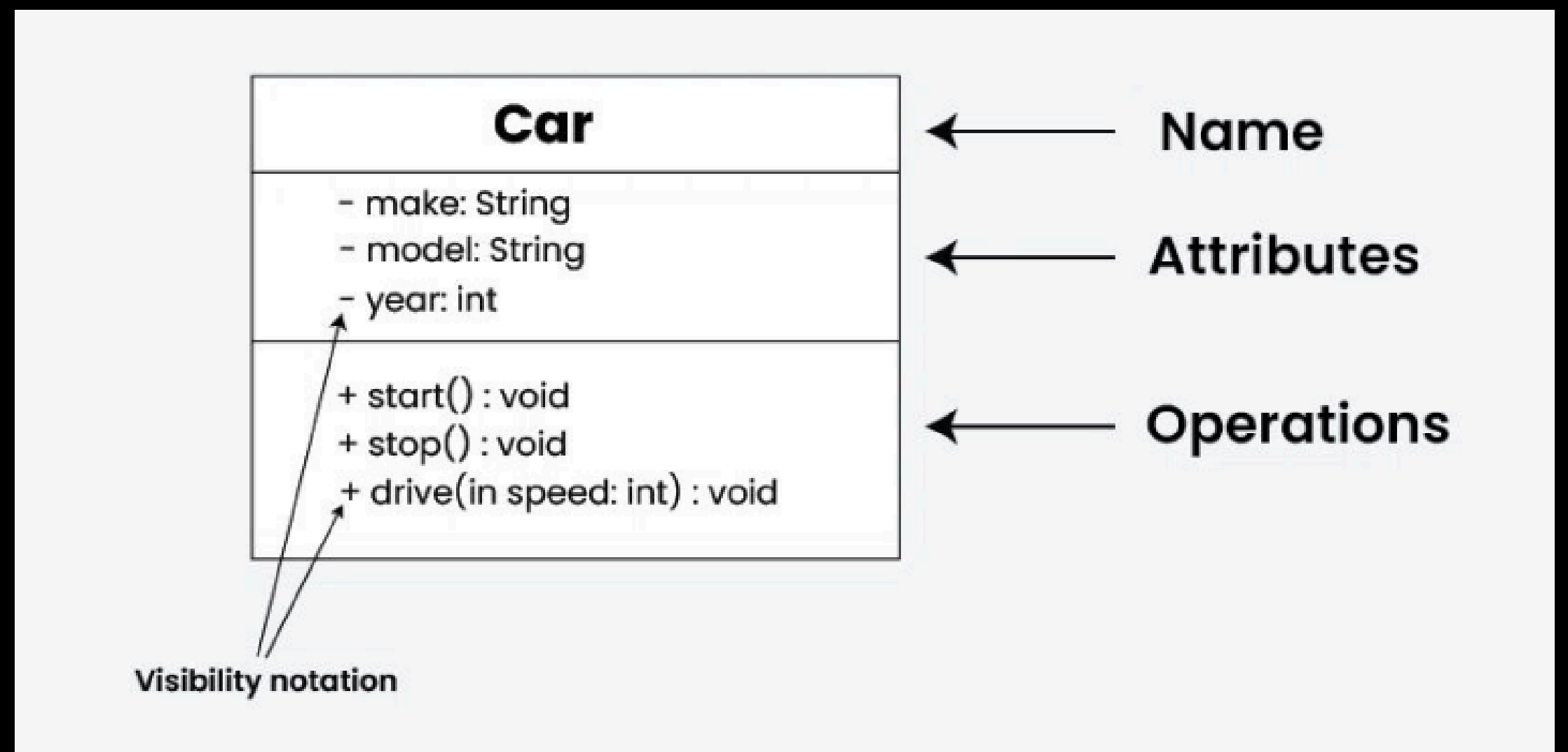
static structure diagram that describes the structure of a system: classes, their attributes, operations and the relationships among objects



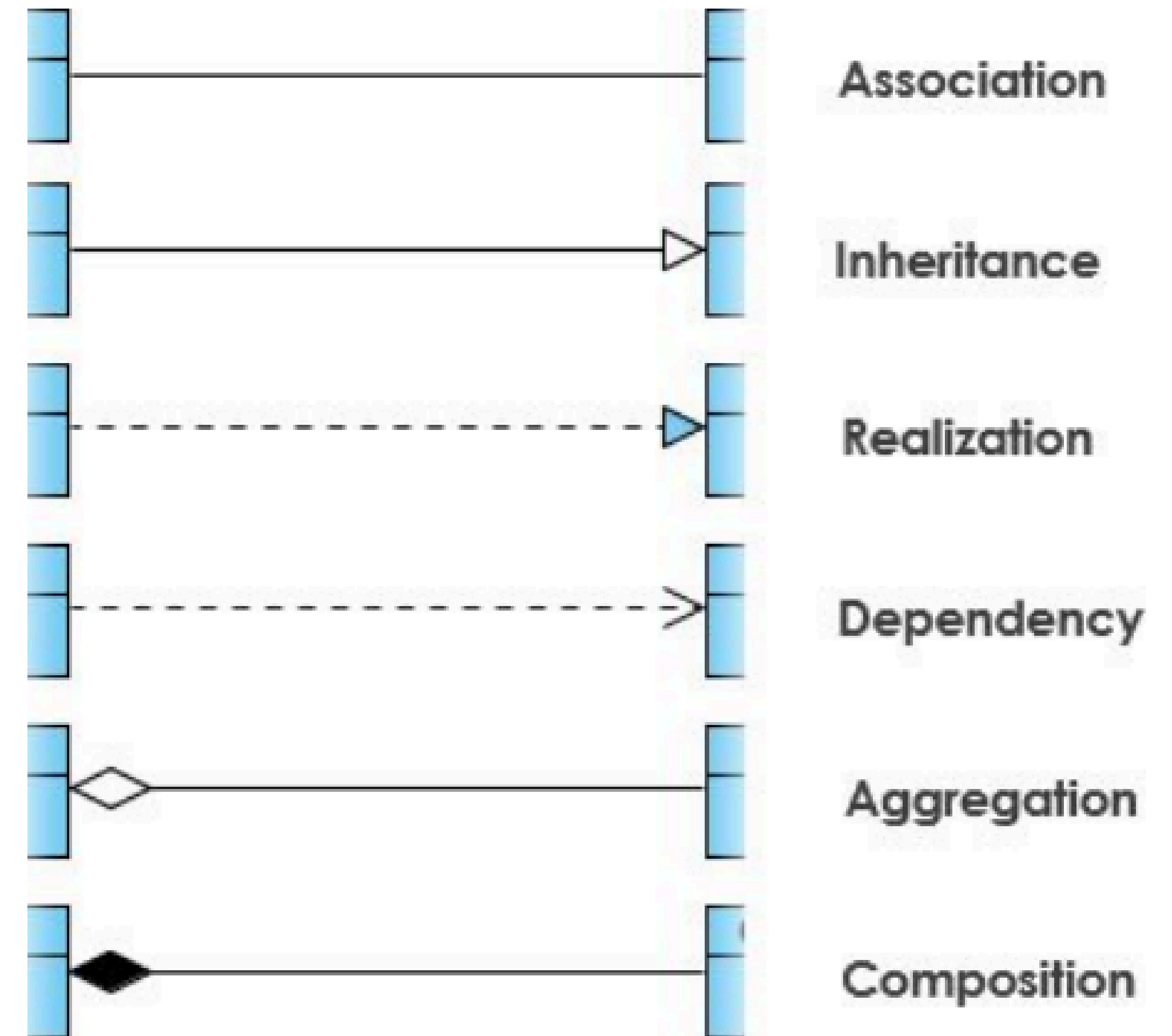


# UML Class Notation

- Name
- Attributes
- Methods / Operations



## Relationship b/w classes + cardinality



# Association

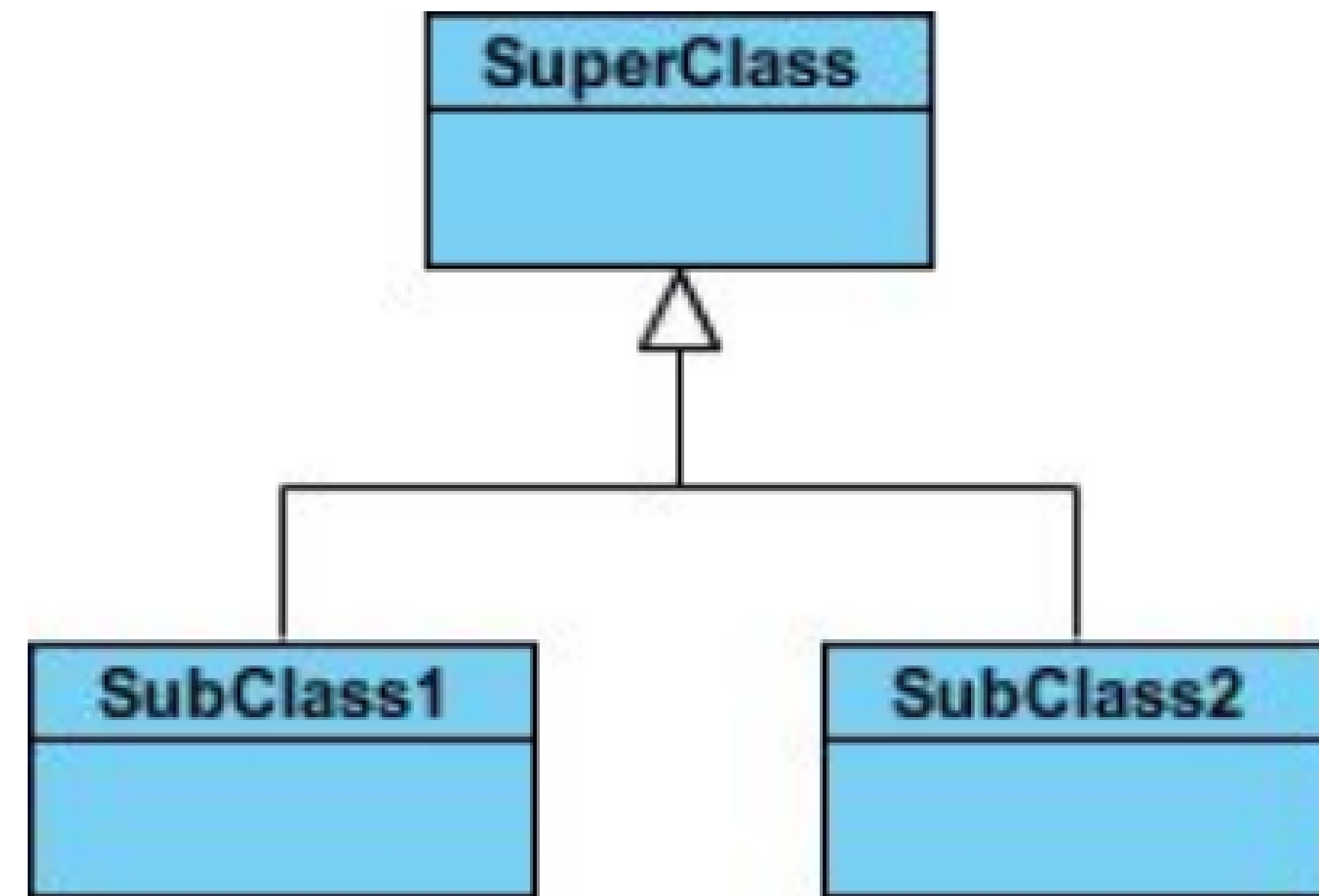
- structural link between two peer classes
- there is an association between Class1 and Class2
- solid line connecting two classes





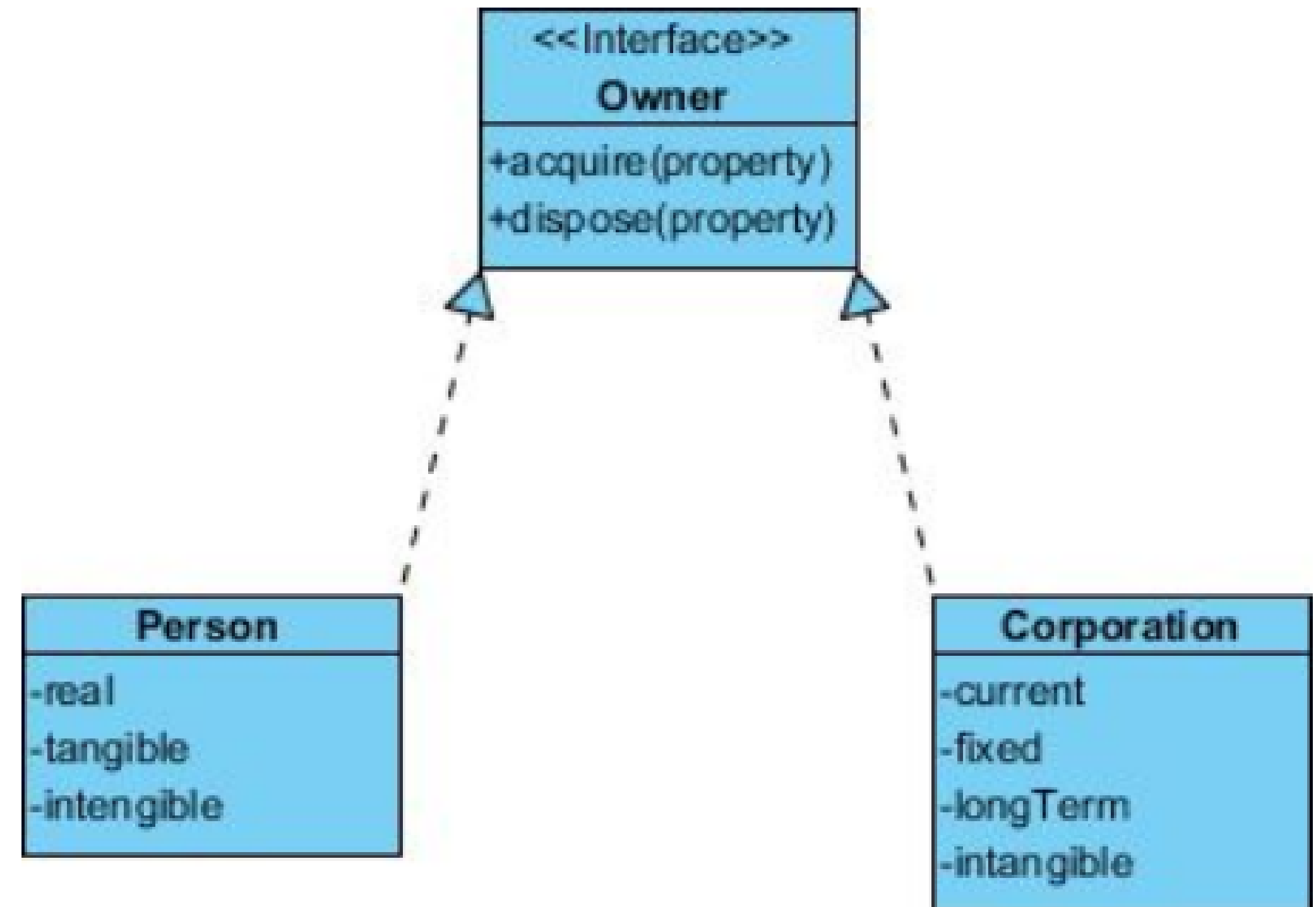
## Inheritance(Generalization)

- represents an "is-a" relationship
- abstract class name is shown in italics
- SubClass1 and SubClass2 are specializations of SuperClass



# Realization

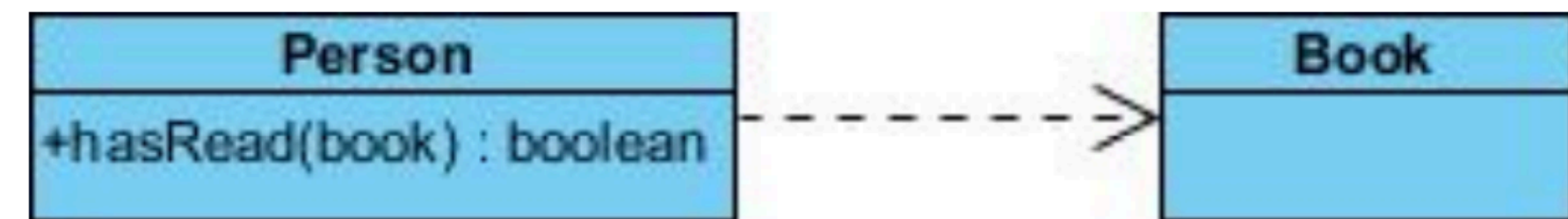
the relationship between the interface and the implementing class





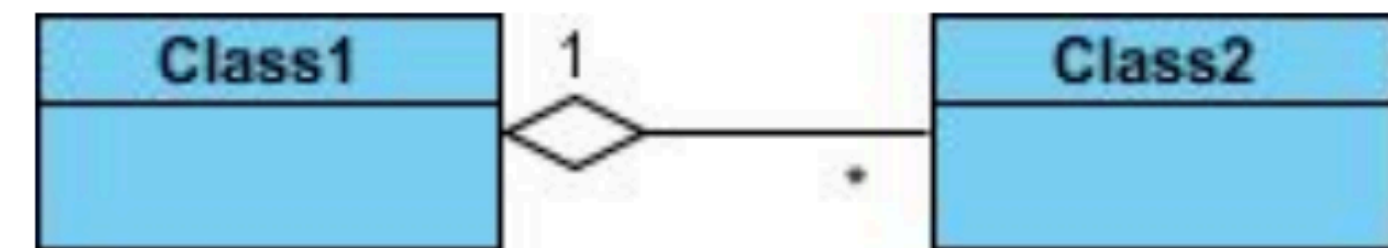
# Dependency

object of one class might use an object of another class in the code of a method



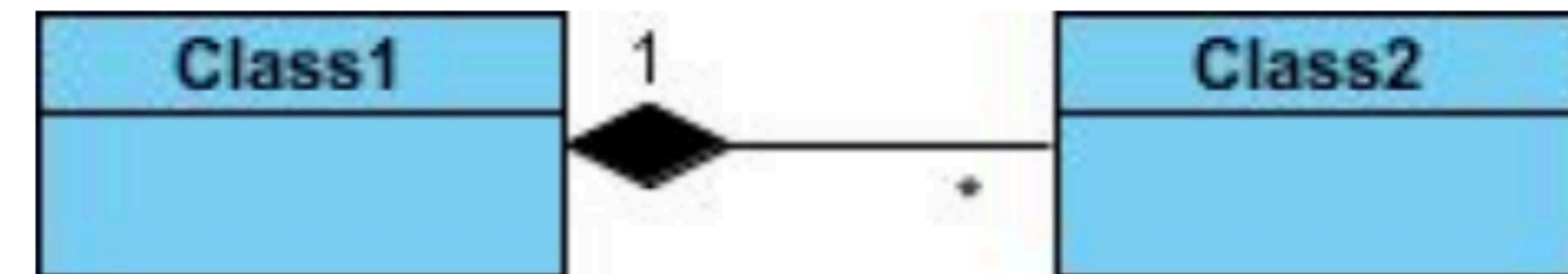
# Aggregation

- many instances of Class2 can be associated with Class1
- objects of Class1 and Class2 have separate lifetimes

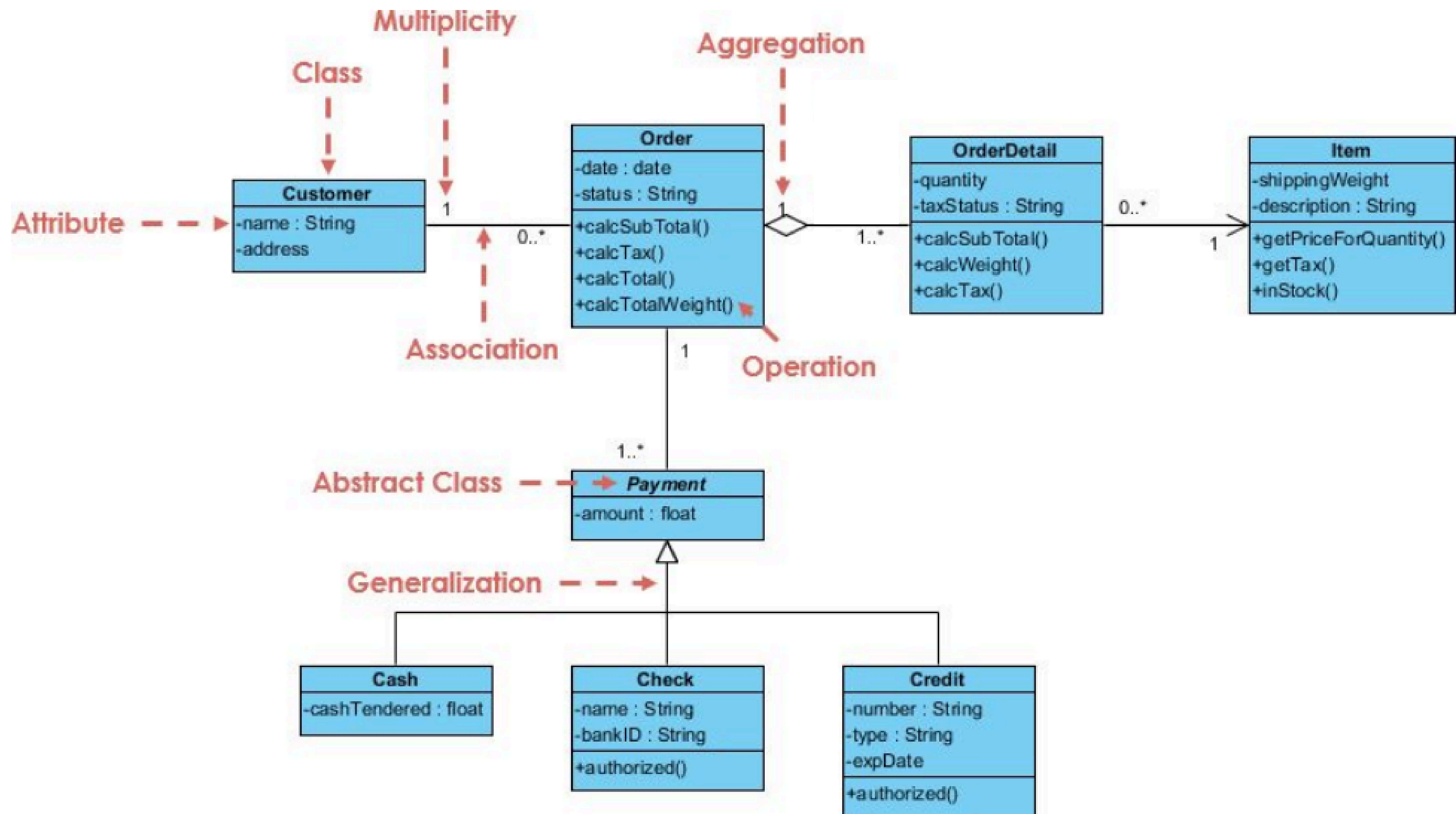


# Composition

- objects of Class2 live and die with Class1
- Class2 cannot stand by itself







UML Class diagram example

# Useful tools

- Visual Paradigm / Sparx Enterprise Architect
- PlantUML / Mermaid